MATH 1271 **Name:** _____
Fall 2026
Quiz 11: 10/20 Content
10/22/2026
Time Limit: 5 minutes **Student ID:** _____

This quiz consists of 1 questions across 2 pages worth 5 points.
Rules:

- You have the indicated amount of time above to complete the quiz and you must not begin working on the quiz until instructed to do so.
- You must show all of your work to receive full credit, unless stated otherwise. **On problems that require work to be shown, answers that are mysterious or unsupported will not receive full credit.**
- You may not use a calculator, textbook, notes or any other external resources.
- Please write your initials at the top of every page in the event the pages become separated.
- If you run out of space on a page, feel free to use the back of the page to continue your work.
- You understand that if you violate any of these or the University's rules, you will receive a score of zero on this quiz and you may be subject to further disciplinary action.

Following the University of Minnesota's policy on academic honesty, I pledge that the work shown on this quiz is entirely my own and that I have followed the rules stated above.

Signature: _____

Question	Points
1	5
Total:	5

1. (5 points) Find the equation of the line tangent to the curve

$$2\sin(x)\cos(y) = 1 \tag{1}$$

at $(\frac{\pi}{4}, \frac{\pi}{4})$.

Solution: We implicitly differentiate first:

$$2\sin(x)\cos(y) = 1 \tag{2}$$

$$\frac{d}{dx}(2\sin(x)\cos(y)) = \frac{d}{dx}1 \tag{3}$$

$$\frac{d}{dx}(2\sin(x)) \cdot \cos(y) + (2\sin(x)) \cdot \frac{d}{dx}\cos(y) = 0 \tag{4}$$

$$2\cos(x)\cos(y) + 2\sin(x)\frac{dy}{dx}\frac{d}{dy}\cos(y) = 0 \tag{5}$$

$$2\cos(x)\cos(y) + 2\sin(x)\frac{dy}{dx} \cdot -\sin(y) = 0 \tag{6}$$

$$\frac{dy}{dx} = \frac{\cos(x)\cos(y)}{\sin(x)\sin(y)} = \cot(x)\cot(y) \tag{7}$$

which at $(\frac{\pi}{4}, \frac{\pi}{4})$ is $1 \cdot 1 = 1$. We then have the tangent line as

$$y - \frac{\pi}{4} = 1 \cdot (x - \frac{\pi}{4}) \tag{8}$$